

SMART ATTENDANCE SYSTEM USING FACIAL RECOGNITION AND FACENET

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ABSTRACT

This paper presents a real-time smart attendance system using FaceNet embeddings and SVM classification deployed on Raspberry Pi 4, achieving 96.2% recognition accuracy under controlled conditions.

INTRODUCTION

Proxy attendance and manual recording errors are persistent problems in higher education institutions. Biometric-based systems using fingerprint scanners have been adopted but require physical contact. Facial recognition offers a contactless alternative. This study proposes a FaceNet-based system integrated with a web dashboard for real-time attendance logging, targeting 200 students at Cavite State University Imus Campus.

METHODOLOGY

FaceNet (Inception ResNet V1) was employed to generate 128-dimensional face embeddings. An SVM classifier with an RBF kernel was trained on a dataset of 2,000 face images (200 students $\times$ 10 images). Images were captured under three lighting conditions. The system was deployed on Raspberry Pi 4 (4GB RAM) with a Logitech C920 webcam. Evaluation metrics: accuracy, False Acceptance Rate (FAR), False Rejection Rate (FRR), and average processing time per recognition event.

RESULTS

The system achieved 96.2% recognition accuracy under controlled lighting. FAR was 0.3% and FRR was 3.5%. Average processing time was 350ms per recognition event, suitable for real-time use. Performance dropped to 88% under low-light conditions. The web dashboard correctly logged attendance events with 99.1% database write success rate.

DISCUSSION

FaceNet-based recognition meets practical accuracy requirements for attendance systems. Low-light sensitivity is the primary limitation, attributed to insufficient IR-compensated training data. Raspberry Pi 4 computational limits become apparent at >5 simultaneous recognition events. Future work should incorporate infrared cameras and multi-face detection for classroom deployment.